

CHAPTER 7

Oil Product Markets

What a barrel becomes — and how refined fuels are priced and traded.

● Gasoline

● Diesel

● Jet

● Fuel oil



Crude is only the raw material

42

gallons in a barrel

Each barrel is split into a slate of refined products — and each fuel trades in a market of its own.

A barrel is a bundle

No one consumes crude oil directly — its value comes entirely from the products refined out of it.

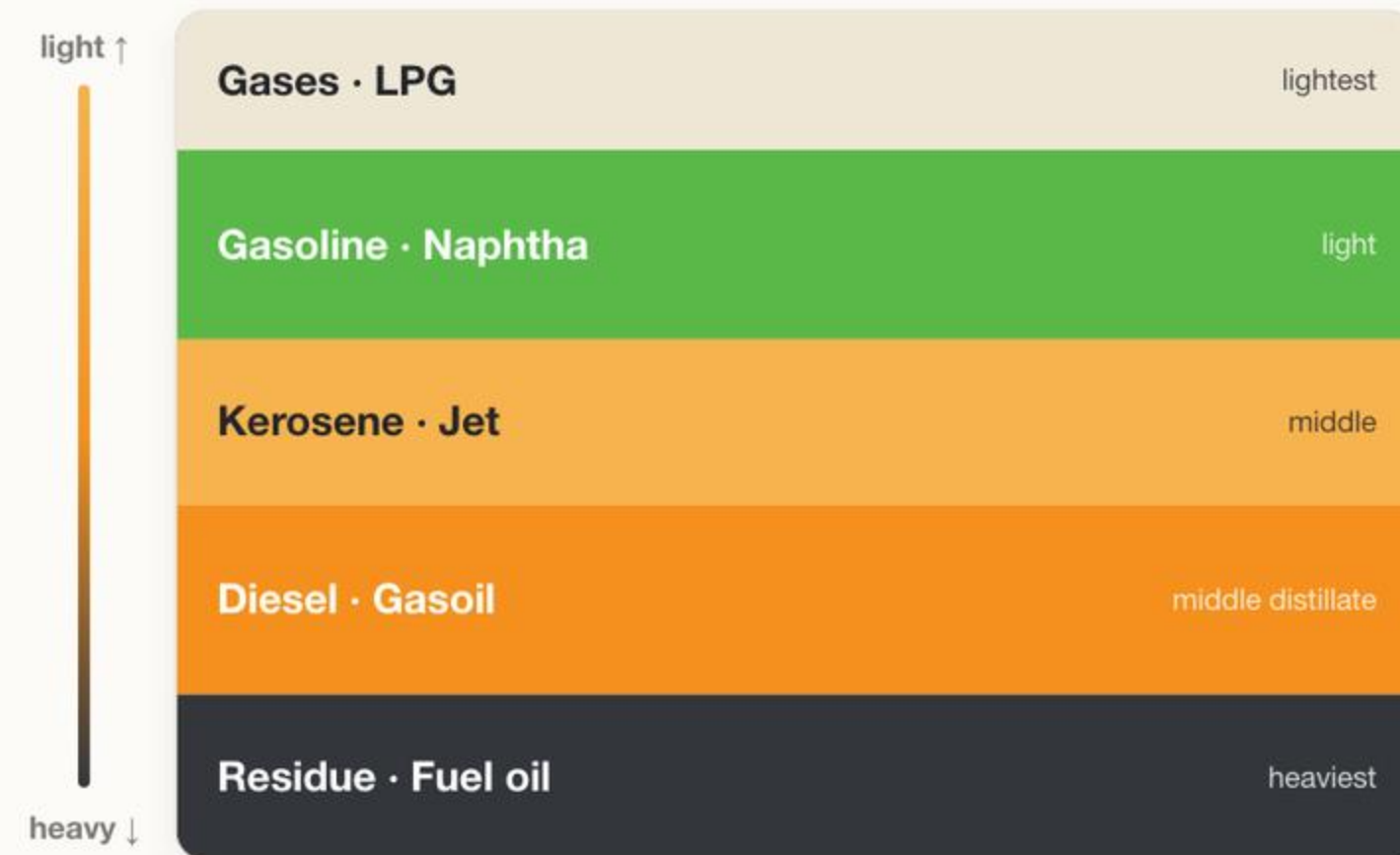
Products are what we buy

The fuel at the pump, in the jet and in the truck is a refined product, priced separately from the crude it came from.

Two prices, one barrel

Crude is the input cost; products are the revenue. The gap between them decides whether a refinery runs at all.

From crude to products



Distillation first

Crude is heated in a tower and separated by boiling point — light gases at the top, heavy residue at the bottom.

Light, middle, heavy

Cuts group into families. The middle distillates — diesel and jet — sit at the heart of industrial and transport demand.

Conversion adds value

Cokers and crackers turn cheap heavy residue into valuable light products, lifting the worth of every barrel.

What comes out of a barrel

TYPICAL REFINERY YIELD · PER BARREL



Gasoline

The single largest product, fueling the world's cars — demand peaks during the northern-hemisphere summer.

Diesel & gasoil

The workhorse fuel of trucks, trains, farms and industry — its demand tracks the real economy more closely than any other.

Jet fuel (kerosene)

Tied to air travel and closely related to diesel — among the products most sensitive to the health of the global economy.

Fuel oil, LPG & naphtha

Heavy fuel oil powers ships; LPG heats and cooks; naphtha feeds the petrochemical industry that makes plastics.

The crack spread



The core idea

A refiner's profit is the gap between what it pays for crude and what it earns selling products — that gap is the crack spread.

The 3-2-1

The standard benchmark — three crude yield two gasoline and one distillate — the most-watched gauge of refining health.

Why it matters

A wide crack pushes refiners to run flat out; a thin or negative crack forces run cuts and tightens product supply.

It can be traded

Refiners hedge their margin with crack-spread futures on the CME, passing price risk to the financial market.

Gasoline

The benchmark

In the US the futures contract is RBOB gasoline, traded on NYMEX and quoted in cents per gallon.

Octane & blending

Gasoline is a blend tuned to octane and emissions specs that vary by region and season — creating distinct local markets.

Seasonality rules

Demand and price climb into the summer driving season and ease in winter, reinforced by seasonal fuel-blend rules.

Sensitive to the consumer

As the fuel people buy at the pump, gasoline prices feed straight into household budgets and into politics.

Diesel

The industrial barometer

Diesel and gasoil move freight, run machinery and power generators — when the economy accelerates, distillate demand leads.

Structurally tight

Harder to make and harder to substitute than gasoline, so its market can squeeze sharply when supply is disrupted.

Global, not local

Diesel trades across oceans between regions, so a shortage in one place is quickly felt everywhere.

~\$83

record diesel crack, 2022

After Europe lost Russian diesel, cracks spiked to a record — a reminder of how violently this market can tighten.

The lighter and heavier ends

Jet fuel

A close cousin of diesel, priced off the same middle-distillate pool — its demand swings with air travel and the travel season.

The IMO 2020 shift

Tighter marine-fuel sulfur limits pushed shipping toward cleaner fuels, permanently widening the value gap between light and heavy.

Fuel oil

The heavy residue at the bottom of the barrel, burned by ships and some power plants — low-margin, but part of every slate.

LPG & naphtha

The lightest cuts feed heating, cooking and the petrochemical plants that turn oil into plastics and chemicals.

Where products are priced

United States

RBOB · ULSD (NYMEX)

US Gulf Coast is the physical pricing center for gasoline and diesel.

Europe

ICE low-sulfur gasoil

The diesel benchmark, anchored on the Northwest Europe (ARA) hub.

Asia

Singapore

The pricing hub for the fastest-growing product market, the marker for the region.

↔ **Arbitrage links them.** When one region's product price runs ahead, cargoes flow in to capture the spread, tying the global market together.

DRIVERS

What moves product prices



Crude, with a lag

Products follow crude but not instantly — after a spike, refiners selling at new prices briefly capture a wider margin.



Seasonality

Gasoline strengthens through summer, distillates through winter — each product runs on its own calendar.



Refining capacity

Unplanned outages, run cuts and seasonal maintenance turnarounds tighten supply and lift cracks.



Regulation & quality

Sulfur limits, blend mandates and biofuel rules reshape which products are valuable and where they can be sold.

When crude spiked, the pump felt it

\$3.99

US retail gasoline, /gal

End of March 2026

\$5.40

US retail diesel, /gal

End of March 2026

~\$28

gasoline crack, /bbl

Scarce barrels rationed by price

Spike and lag

When Brent surged in early 2026, product prices followed with a delay, briefly handing refiners unusually wide margins.

Easing into mid-year

As crude retreated below \$100 and Hormuz traffic recovered, product prices and cracks began to normalize.

CONCLUSION — OIL MEETS THE CONSUMER

Crude is only valuable for what it becomes.

Products are the point

The product markets are where oil finally meets the consumer.

The crack is the link

The spread between crude and products governs how hard refiners run and how much fuel reaches the market.

Each is its own market

Gasoline, diesel and jet each have distinct demand, seasonality and benchmarks — yet all move with crude.